



UNIVERSITY OF
OREGON

Single-Cell Analysis of JAK/STAT Inhibitors on IL-6 Signaling



Sean Bergan^{1*}, Catherine Carmosino^{1*}, Joyce Wang^{1*}, Stephanie D. Anover-Sombke², Katherine Brower, Alexandra C. Ferrannini², Jessica Garber², Palak C. Genge², Lucas T. Graybuck², Yudong He², Alexander T. Heubeck², Neelima Inala², Garth L. Kong², Christian M. LaFrance², Alina Ott², Charlie Puente-Matos², Mansi Singh², Peter J. Skene², Troy R. Torgerson², Peter J. Wittig²

¹Bioinformatics & Genomics Track, KCGIP, University of Oregon, Eugene, OR. ²Allen Institute of Immunology, Seattle, WA.

*Presenting Author, Advisor

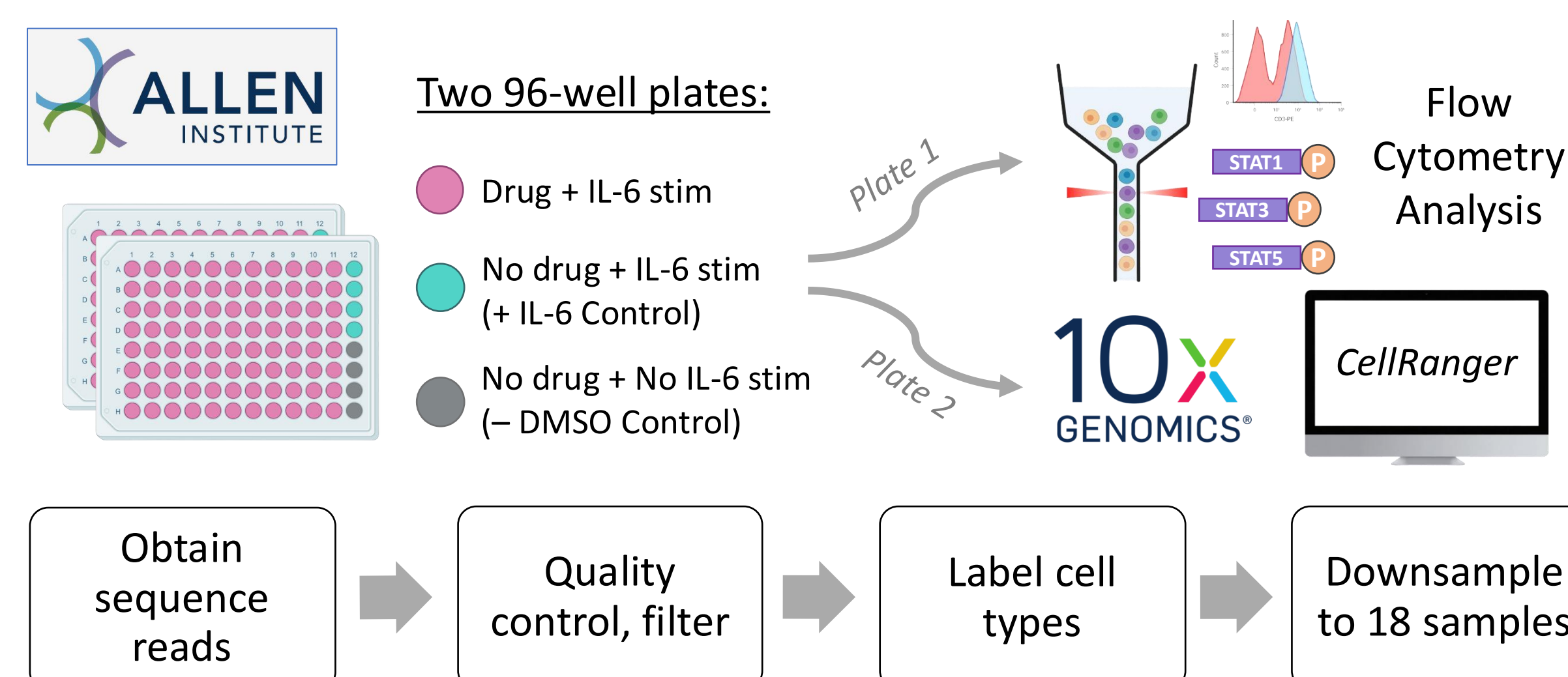
1. Overactive IL-6 and Inflammation

The Interleukin 6 (IL-6) JAK/STAT pathway helps regulate the body's **inflammatory response**. The **IL-6 cytokine** binds to cell surfaces and activates **JAK/STAT proteins** downstream, turning on and off transcription factors. **Naive T-cells** have IL-6 receptors and are IL-6 responsive, whereas **effector memory T-cells** lack IL-6 receptors and are IL-6 nonresponsive. High levels of IL-6 are part of **cytokine storm** responses in diseases like COVID-19 and autoimmune diseases such as rheumatoid arthritis. Common treatments of these diseases are **inhibitors along the JAK/STAT pathway**.

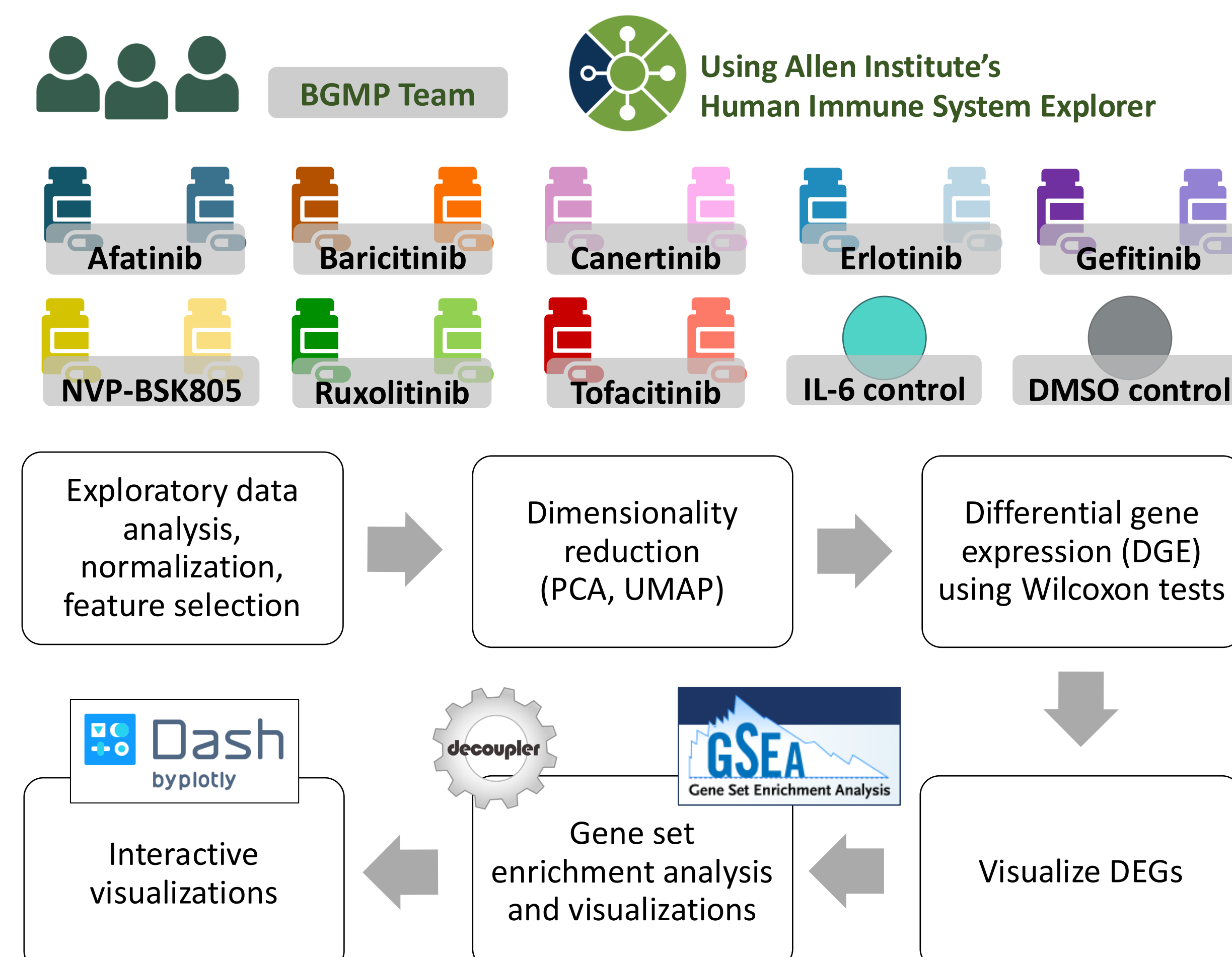
This project aims to investigate isolated T-cell response to 16 JAK/STAT inhibitors, identify differences between formulations, and explore off-target effects using single-cell RNA-sequencing data.

2. Methods and Workflow

EXPERIMENTAL METHODS AND PREPROCESSING



DATA ANALYSIS



3. Eight Cell Types and 18 Drug Treatments

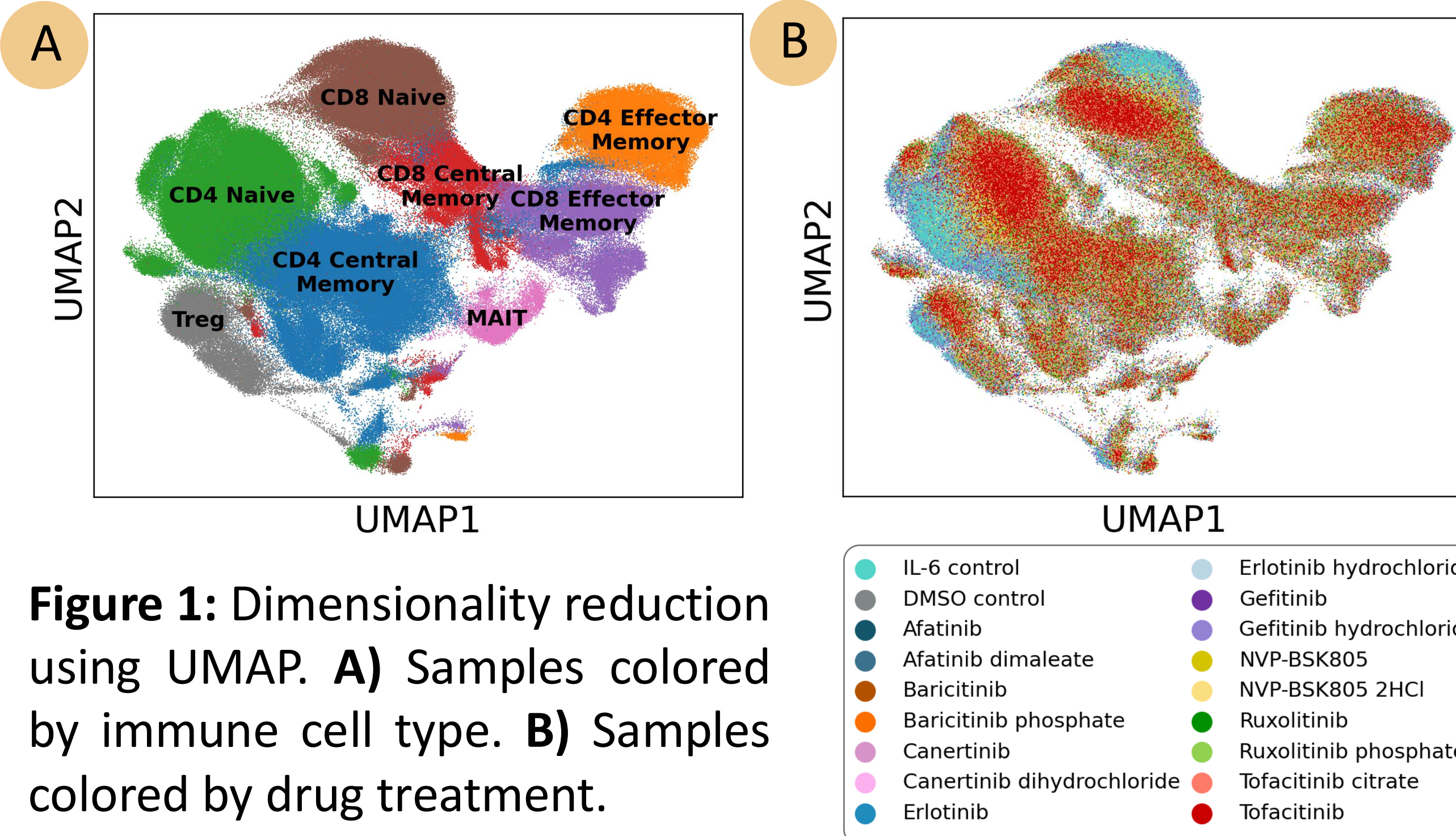


Figure 1: Dimensionality reduction using UMAP. **A)** Samples colored by immune cell type. **B)** Samples colored by drug treatment.

Samples colored by the eight immune cell types show clear grouping while samples colored by the 18 drug treatments (including controls) show no clear grouping.

4. DEGs Show Drug Similarity to Controls

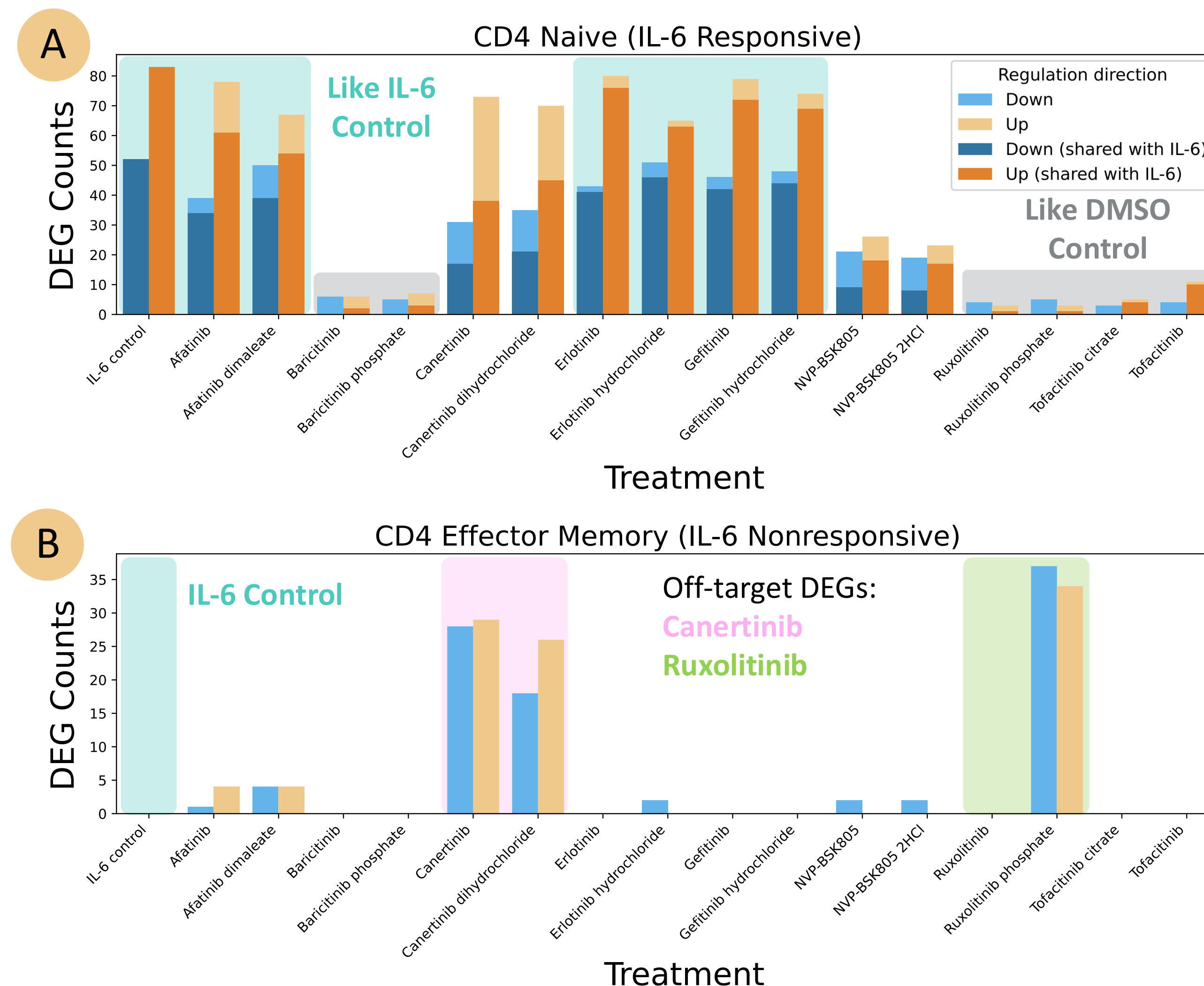


Figure 2: Significant (>1 Log Fold Change, adj. p < 0.05) DEG counts up- or down-regulated compared to DMSO control in **A)** CD4 naive cells (darker colors represent DEGs expressed the same direction as IL-6) and **B)** in CD4 effector memory cells.

Higher number of genes that have similar expression to the DMSO control point to drugs which may be more effective at suppressing IL-6. Some off-target effects are seen with Canertinib (in both cell types) and Ruxolitinib phosphate (only in CD4 effector memory cells).

5. GSEA Demonstrates Off-Target Effects

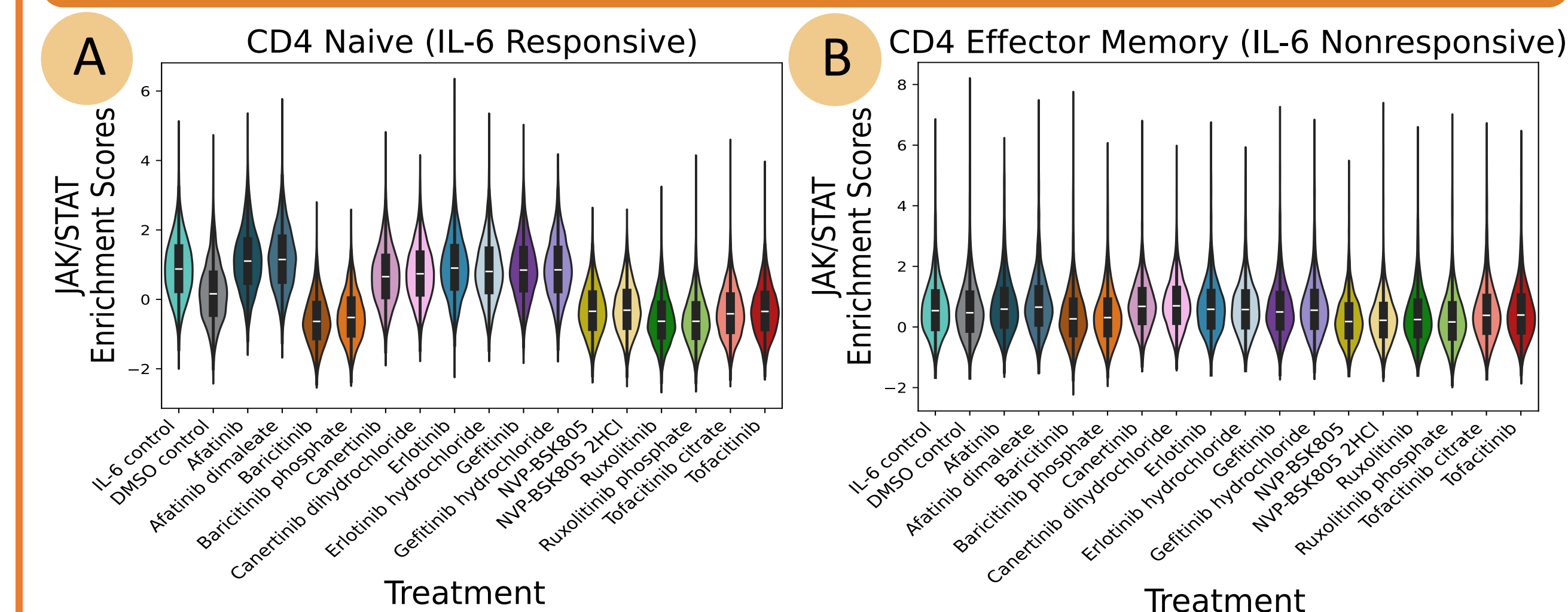


Figure 3: Violin plots of JAK/STAT pathway enrichment scores from the PROGENy database. **A)** Expression in CD4 naive cells. **B)** Expression in CD4 effector memory cells.

Certain inhibitors reduced expression of the JAK/STAT pathway only in naive cells.

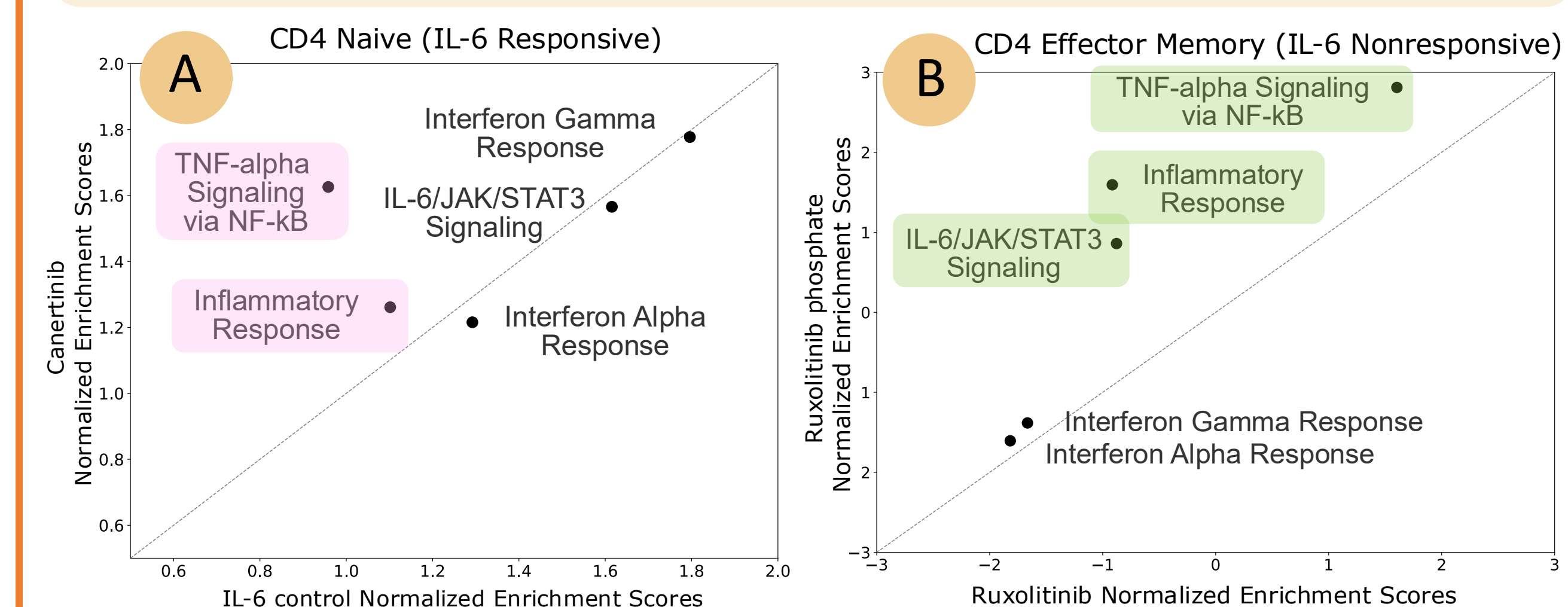


Figure 4: Scatter plots of normalized enrichment scores for five hallmark pathways. **A)** Canertinib vs. IL-6 control in CD4 naive cells. **B)** Ruxolitinib formulations in CD4 effector memory cells.

Two pathways show off-target enrichment in Canertinib-treated cells and three pathways show formulation-dependent differences in Ruxolitinib-treated cells.

6. Conclusions and Future Directions

Conclusions:

- Strong inhibitors:** Baricitinib, NVP-BSK805, Ruxolitinib, Tofacitinib
- Weaker inhibitors:** Afatinib, Erlotinib, Gefitinib
- Off-target effects** in Canertinib-treated cells
- Differences within** Ruxolitinib formulations

Future Directions:

- Continue to develop interactive visual tools and certification of reproducibility.

7. Acknowledgements and References

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